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Exploring the future of privacy-preserving heart disease prediction: a fully homomorphic encryption-driven logistic regression approach

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Abstract

Homomorphic Encryption (HE) offers a revolutionary cryptographic approach to safeguarding privacy in machine learning (ML), especially in processing sensitive healthcare data. This study aims to address the critical issue of privacy-preserving heart disease prediction by developing a novel Homomorphic Encryption-Driven Logistic Regression (HELR) framework, leveraging the Cheon-Kim-Kim-Song (CKKS) encryption scheme. The framework was implemented using the TenSeal and Torch libraries and evaluated on encrypted heart disease datasets with varying polynomial degrees. The study's design involved applying the HELR model to three healthcare datasets and comparing its performance with Support Vector Machines (SVM). The major findings revealed that the HELR model achieved high accuracy, within 1% to 3% of its non-HE counterpart, while maintaining competitive computational efficiency. Furthermore, the HELR framework demonstrated robust security against various privacy attacks, including poisoning, evasion, member inference, model inversion, and model extraction, at different ML stages. Notably, the HELR model outperformed SVM in terms of accuracy, showcasing its effectiveness for secure healthcare predictions. The results suggest that HE-enhanced models can offer secure, accurate predictions, paving the way for advancements in privacy-preserving healthcare analytics. However, the study identified limitations related to the computational overhead introduced by HE and the scalability of the model for large datasets. Future work will focus on optimizing encryption techniques and exploring parallel processing methods to address these challenges.

Keywords: CKKS, Gradient Descent, Healthcare, Heart disease prediction, Homomorphic Encryption, Logistic Regression

Introduction

Machine learning (ML) has attracted considerable attention because of its ability to address real-world challenges across diverse domains such as artificial intelligence, bioinformatics, medical sciences, and marketing. ML can extract valuable insights from extensive datasets without requiring deep domain-specific expertise. However, the healthcare industry faces a significant hurdle in safeguarding patient privacy and sensitive medical information.

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Homomorphic Encryption (HE) (Salim (2022)) [16] has emerged as a compelling solution for addressing this concern. HE encryption technique permits specific mathematical operations to be executed on encrypted data without decryption. This capability enables computations to be performed directly on encrypted data, with the results remaining encrypted. This feature is particularly advantageous when sensitive data must be processed in untrusted environments or by third parties, to ensure privacy and security.

Traditional privacy-preserving techniques in ML [17, 18, 26], such as anonymization, perturbation, randomization, and condensation, aim to protect privacy but often incur information loss and reduce data utility. Differential privacy, while practical, introduces noise that may compromise the accuracy of the ML models. In contrast, combining HE with logistic regression Zhou [23], Wei [24], Duchi [2], Machanavajjhala [1] facilitates privacy-preserving computations without sacrificing data utility or accuracy. Gupta, R. et al. [30] Introduces a quantum machine learning-based model to predict malicious users in cloud networks, leveraging quantum algorithms to enhance anomaly detection and bolster network security. Gupta, R. et al. [31], Proposes a framework for securing IoT systems integrated with cloud services, addressing data privacy and integrity challenges through innovative security mechanisms. Later in the same year Gupta, R. et al. [32], Develops a privacy-preserving model for medical data using differential privacy and tri phase adaptive learning, ensuring data protection while maintaining utility and accuracy. In addition to it, Gupta, R. et al. [33], Presents a machine learning model incorporating differential privacy for secure data classification in cloud environments, suitable for privacy-sensitive applications. Gupta, R. et al. [34], Introduces a differential privacy-based model for handling sensitive data in multimedia and application-driven contexts, ensuring confidentiality without sacrificing usability.

Given the drawbacks of differential privacy and traditional methods, a framework that integrates HE with logistic regression is essential. The CKKS scheme, a variant of HE tailored for real-number operations, supports batch processing and efficient handling of large datasets. This makes it well-suited for constructing logistic regression models using encrypted healthcare data, ensuring that patient privacy is maintained throughout the process.

The motivation for this research is to address the pressing need for privacy-preserving heart disease prediction systems that balance robust data security with predictive accuracy. By leveraging FHE, the proposed approach ensures that patient data remains secure throughout the computational process, fostering trust in predictive analytics systems while adhering to stringent privacy regulations. This study aims to advance the future of secure healthcare analytics by demonstrating the feasibility and effectiveness of a fully homomorphic encryption-driven logistic regression model for heart disease prediction.

In healthcare, HE holds promise despite its limited application in practical ML tasks, particularly in integration with logistic regression (Wibowo [19], Yu [20]) models. Logistic regression is a widely adopted ML method for classification, owing to its simplicity and competitive prediction accuracy. It is a foundational element in more complex classification algorithms such as gradient boosting. Logistic regression is a relatively lightweight algorithm suitable for execution on mobile devices with limited computational resources. Mobile devices can participate in federated learning by locally training logistic

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regression models on their own data and sending updates to a central server for aggregation, this study proposes a framework for an HE-aware logistic regression model using the CKKS scheme to predict medical outcomes based on patient data. For instance, the model can forecast the probability of a patient developing a specific medical condition based on their medical history and pertinent factors. Training the model on encrypted healthcare data guarantees that patient privacy remains protected during each stage of analysis.

Contributions

The key contributions of this study are:

1. *HE-Driven Logistic Regression Model* A logistic regression model leveraging CKKS homomorphic encryption was developed to ensure data privacy without compromising accuracy.
2. *Robust Security Analysis* The system demonstrated strong resilience against threats like poisoning, evasion, membership inference, model inversion, and extraction attacks.
3. *Effective Performance Evaluation* Tested on Kaggle healthcare datasets using TenSEAL, the model maintained high accuracy and practicality while securing sensitive data.

The paper is structured as follows: Sect. "RELATED WORKS" provides a comprehensive review of prior research and studies related to the topic, highlighting existing approaches and identifying gaps that the current work addresses. Sect. "BACKGROUND KNOWLEDGE" lays the groundwork for the study by explaining the underlying concepts, theories, or technologies essential for understanding the proposed approach. Sect. "PROPOSED SYSTEM" outlines the methodology, detailing the techniques, processes, and frameworks used to achieve the study's objectives. Sect. "SECURITY ANALYSIS" evaluates the robustness and effectiveness of the proposed method through a security analysis, examining vulnerabilities and demonstrating resilience against potential threats. Finally, Sect. "EXPERMENTATION AND RESULTS" concludes the paper by summarizing the key findings, contributions, final section discusses the conclusion, limitations and suggesting directions for future research.

Related works

Improving the precision of health data predictions while safeguarding privacy through Homomorphic Encryption (HE) (Paul [11], Scheibner [15]) and Machine Learning (ML) integrates established ML methodologies and offers practical and scholarly advantages. Numerous studies introduced innovative methods to address the dual challenges of data precision and confidentiality. Chandramohan, D, et al. [5] proposed a hybrid authentication technique to enhance privacy preservation in cloud storage, ensuring access only to authorized users while mitigating attacks. Their approach demonstrated improved utility and security for sensitive user data in vulnerable cloud environments.

Huang et al. [7] implemented a federated learning strategy by training models on decentralized electronic medical records (EMRs) from multiple hospitals. This approach

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ensures accurate predictions by utilizing diverse datasets, without requiring data sharing, thereby bolstering data privacy. The efficacy of federated learning in healthcare underscores its potential to uphold data accuracy, while mitigating privacy concerns. This research provides a foundational framework for future inquiries into decentralized ML models. Similarly, Kim et al. [25], Abadi M [3], Abadi M [4] combined differential privacy and homomorphic encryption in logistic regression to achieve secure and efficient model building, thereby demonstrating a dual-layered ML data security approach.

Fan et al. [8] introduced algorithms to facilitate the analysis of large-scale health data while upholding stringent privacy and security standards. Their contributions have significantly advanced privacy-preserving techniques for big data, laying a solid foundation for future improvements in algorithmic efficiency and privacy protection. Boura et al. [9] proposed a hybrid solution that integrates TFHE, B/FV, and HEAAN schemes within Fully Homomorphic Encryption (FHE) to enhance data privacy and ensure secure data processing. This approach effectively combines multiple HE schemes by leveraging their respective strengths, while addressing individual limitations.

Guo et al. [12] developed a mutual privacy-preserving k-means strategy based on homomorphic encryption, safeguarding participant privacy and cluster center data. Their study demonstrated the application of HE to clustering algorithms, offering insights into the preservation of privacy in unsupervised learning tasks. Wood et al. [13] provided a comprehensive overview of Fully Homomorphic Encryption (FHE) and its applications in medicine and bioinformatics, informing practitioners about state-of-the-art HE techniques. Their critical review identified research gaps and opportunities for applying FHE to medical and bioinformatic data.

Li et al. [18] introduced a novel homomorphic encryption framework over non-abelian rings, achieving one-way security based on the Conjugacy Search Problem. Their work contributes to new theoretical foundations for HE by expanding secure encryption methods in computational mathematics. Carpov et al. [21], Ghavamipour [22] advanced the practical application of HE in logistic regression by employing semi-parallel training using the BFV cryptosystem and FHE on encrypted data. Ren et al. [10] presented a secure homomorphic encryption scheme for data operations in Internet of Things (IoT) systems, ensuring critical data privacy. Their study explored the intersection of IoT and HE, paving the way for secure data processing in increasingly connected environments. Chen et al. [6] utilized bibliometric analyses to uncover trends in machine learning research within higher education, providing insights into prevalent research themes and collaborations.

Wei et al. [24] proposed a privacy-preserving two-party logistic regression algorithm for vertically partitioned data, using asynchronous gradient sharing to enhance secure collaborative learning. Their research highlights the potential of asynchronous methods in privacy-preserving ML, thereby contributing to more efficient collaborative learning frameworks. Sarkar et al. [14] explored privacy-preserving genotype imputation using machine learning and Paillier cryptosystem-based homomorphic encryption, demonstrating HE versatility in secure genetic data analysis.

Despite the promise of homomorphic encryption (HE) in healthcare, its application in practical machine learning (ML) tasks, particularly in logistic regression models, remains limited. Traditional privacy-preserving ML techniques such as anonymization,

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perturbation, randomization, and condensation often result in information loss and reduced data utility. Combining HE with logistic regression allows privacy-preserving computations without sacrificing data utility or accuracy.

Aono et al. [28] discussed scalable and secure logistic regression using homomorphic encryption, focusing on the efficiency and practicality of the approach. Reviewing the methodology, optimization, and performance benchmarks is crucial. Chen et al. [29] explored logistic regression over encrypted data using fully homomorphic encryption, with applications in medical genomics. Key aspects to review include encryption schemes, handling large datasets, and real-world scenarios.

The proposed framework integrates CKKS-based HE with logistic regression to predict medical outcomes, while preserving patient privacy. This approach addresses the gaps identified in the existing methods by ensuring data utility without compromising privacy. The framework is adaptable to other ML techniques, such as Support Vector Machines (SVM), deep learning, and gradient boosting, offering a foundation for comprehensive evaluations across diverse datasets. By leveraging the strengths of the CKKS-based HE, the proposed method enhances the security and efficiency of privacy-preserving computations. This integration not only maintains the accuracy of health data predictions but also ensures robust privacy protection, making it a valuable contribution to the field of secure and efficient machine learning in healthcare.

Background knowledge

This section presents symbols.

Notations

Table 1 lists the symbols and their descriptions used in the subsequent sections.

Table 1 List of symbols and notations used in this study

Symbol	Description
$\varphi(M)$	M^{th} cyclomatic polynomial $\varphi_M(X)$
N	Degree of cyclomatic polynomial $\varphi_M(X)$
R	Ring of the polynomial defined for plaintext space
R_q	Residue ring of the polynomial defined for ciphertext space
H	Subspace of C^N
γ	Canonical embedding
π:	Mapping function, which projects a vector from a sunspace C^N to $C^{N/2}$
P	No of Records
D	No of features
W	Encrypted weight vector
σ	Sigmoid function
α	Learning rate
Δ	Scaling factor
(c_1, c_2)	Ciphertext
z	Medical record
Tar	Classification Result
m	Encoded medical record
E, D	Encryption and Decryption
PR_k, PU_k	User k's Private and public keys

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Logistic regression

Logistic regression is a statistical technique used to predict the likelihood of a binary outcome based on one or more explanatory variables. This method is frequently used in classification problems, where the objective is to estimate the probability of an instance belonging to a specific group.

The core principle of logistic regression involves examining the relationship between independent variables (also known as predictors or features) and the probability of an outcome falling into a particular category. Unlike linear regression, which forecasts continuous outcomes, logistic regression uses a logistic function to model the probability of binary results. The mathematical representation of the logistic regression model can be expressed

$$P(X) = \frac{1}{1 + e^{(\beta_0 + \beta_1 x_1 + \text{----------} + \beta_n x_n}} \quad (1)$$

where:

- $P\ (Y=1|X)$ is the probability that the outcome variable Y equals 1, given the input feature X.
- e is the base natural logarithm.
- $\beta_1, \beta_2, \beta_3, \beta_n$ are the coefficients (parameters) associated with each independent variable X_1, $X_2, X_3, \ldots, X_n$ respectively.
- $X_1, X_2, X_3, \ldots, X_n$ are independent variables.

The logistic function, known as the sigmoid function, transforms a linear combination of independent variables and coefficients into a value between 0 and 1. This ensures that the logistic regression output is a probability representing the likelihood of the observation belonging to the positive class (often coded as 1).

During training, the logistic regression model learns the optimal values for the coefficients by maximizing the likelihood of observing the actual outcome given the input features. This process is typically achieved using optimization algorithms, such as maximum likelihood estimation or gradient descent.

Once the model is trained, it can predict the probability of an observation belonging to a positive class based on its features. A common approach is to set a threshold (e.g., 0.5) such that observations with predicted probabilities above the threshold are classified as belonging to the positive class. By contrast, those below the threshold belong to the negative class.

Cheon-Kim-Kim-Song

CKKS (Cheon-Kim-Kim-Song) homomorphic encryption scheme

The CKKS (Cheon-Kim-Kim-Song) encryption scheme, introduced in 2016 by Kim, Cheon, Kim, and Song, is a ground breaking framework tailored for homomorphic encryption. This revolutionary technique enables computations of encrypted data without prior decryption, playing a pivotal role in scenarios such as cloud computing, where safeguarding data privacy throughout computations is paramount.

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Homomorphic properties

CKKS was meticulously designed to support homomorphic addition and multiplication operations on encrypted data. This capability facilitates the direct execution of intricate mathematical operations on ciphertexts, ultimately producing results identical to those derived from decrypted plaintexts.

Efficiency

CKKS is strategically engineered for efficiency in homomorphic encryption, which generally entails higher computational demands than traditional methods. CKKS strives to optimize the performance of real-world applications, ensuring that the computational overhead is manageable.

Support for approximate arithmetic

Unlike certain homomorphic encryption schemes that strictly adhere to exact arithmetic, CKKS proficiently handles approximate arithmetic. This flexibility is invaluable in applications, such as machine learning, where absolute precision in computations may not be imperative.

Components of CKKS

Parameter generation

CKKS requires the generation of specific parameters tailored to the security requirements and desired computational precision. These parameters intricately define the encryption scheme and the permissible range of achievable computations. Key parameters include:

- *The polynomial modulus degree (n)* defines the degree of the polynomial ring used.
- *Coefficient modulus (q)* a large integer modulus determining the noise budget.
- *Scaling factor (*Δ*)* a constant that scales plaintext values to ensure precision.

Key generation

The CKKS involves the generation of both public and private keys.

- Public key (pk) is used to encrypt data.
- Private key (sk) is used to decrypt data.
- Evaluation key (evk): Facilitates homomorphic operations on ciphertexts.

Encryption

The encryption process in CKKS involves encoding the plaintext into a polynomial representation that is conducive to homomorphic operations. This process includes:

i. *Encoding* Converting a plaintext message into a polynomial in the ring $Z[x]/(x^n + 1)$.

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ii. *Scaling* The polynomial is multiplied by a scaling factor to ensure precision.
iii. *Encryption* Random noise is added, and the public key is used to produce ciphertext.

Mathematically, if m is the plaintext, the ciphertext c is given by:

$$C = Enc(m) = ([\Delta \cdot m] + e) mod q \tag{2}$$

where e is the added noise.

Homomorphic operations:
CKKS supports robust homomorphic operations, specifically, addition and multiplication.

Addition: Given two ciphertexts c_1 and c_2, their homomorphic addition is:

$$C_{sum} = c_1 + c_2 \tag{3}$$

Multiplication:
Given two ciphertexts c_1 and c_2, their homomorphic multiplication requires linearization:

$$C_{prod} = Relin(c_1 \cdot c_2) \tag{4}$$

Decryption:
Upon completing the computations, the ciphertext can be decrypted using the private key to reveal the result in its original plaintext form.

i. *Decryption* A private key is used to convert the ciphertext back into a scaled polynomial.
ii. *Decoding* The polynomial is scaled down and mapped back to the original plaintext space.

Mathematically, if c is the ciphertext, the plaintext $m\prime$ is:

$$m' = \frac{Dec(c)}{\Delta} \tag{5}$$

Comparison of CKKS with other Schemes:

Feature	CKKS	BGV
Number Type	Real/Floating-point (approx.)	Integer/Fixed-point (exact)
Noise Handling	Tolerates small approximations	Exact computation with noise mgmt
Scaling	Rescaling	Modulus switching
Efficiency	Faster for real numbers	Slower, precise for integers
Use Case	ML, AI, signal processing	Cryptography, exact arithmetic

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Proposed system

System architecture

The top-level architecture of privacy-preserving heart disease prediction using a logistic regression approach on an encrypted dataset is shown in Fig. 1, which comprises three main roles: patient (p), hospital (h), and Cloud Service Provider (CSP). The CSP provides remote data storage and computing services for authentic and curious users. Users have vast amounts of health information and use CSP services. The responsibilities of an individual member in the proposed arrangement are as follows.

1. *Patient* He/she is the owner of sensitive data; the patient performs the following operations

   - *Health Data encryption* Initially, the user chooses his/her health information to be sent to the hospital for disease prediction, which is considered a test dataset. Then, he/she encrypts his/her sensitive medical data using the hospital public key and sends cipher medical records to the hospital for disease prediction.
   - *Decryption of predicted value* After receiving the predicted value from the hospital in cipher form, use the patient's private key to decrypt it and learn about the presence of the disease.

2. *Hospital* Owners of the training dataset, which consists of patients' medical records from hospital visits. Hospitals generally maintain medical data after receiving approval from patients coming to the hospital for treatment. The doctors are responsible for the following operations:

   - *Encryption of Training model parameters and training dataset* Doctors use the hospital public key to encrypt the training dataset and model parameters, and send the resultant ciphertext to the cloud, which is used for model building.
   - *Updated training dataset* New medical records are inserted into the existing training dataset after obtaining permission from the patient.
   - *Decryption of predicted result* After receiving the predicted value from the cloud in cipher form, use their private key to decrypt and determine the presence of disease.

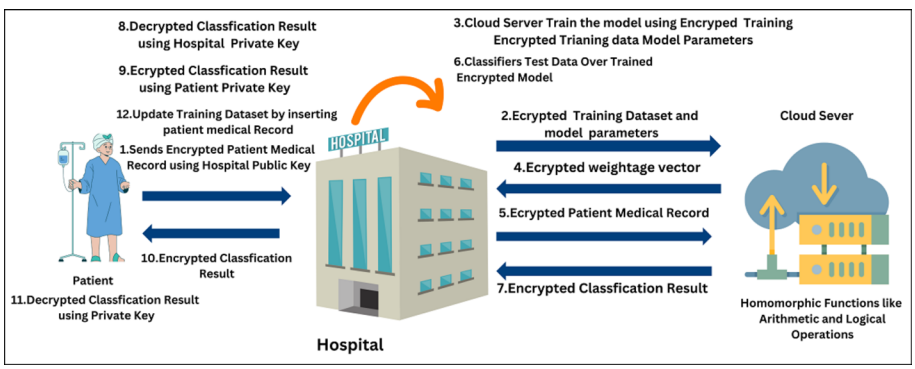


Fig. 1 Top-Level Architecture

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3. CSP, also known as a cloud server, has the following responsibilities of SP include:

   - A gradient-based logistic regression classification model was built for the encrypted training set.
   - The updated weight is sent to the hospital.
   - The test datasets are classified using trained models and the predicted results are returned to the hospital.

Proposed model

The proposed scheme is based on CKKS Fully Homomorphic Encryption and Logistic Regression over encrypted data. It offers an efficient methodology to support privacy at all levels, including the input, output, and model privacy. Whenever a patient visits a hospital based on hospital requirements, a patient securely shares his/her health information by encrypting health records using the hospital's public key, which is considered a test dataset for heart disease prediction. The hospital uses CSP services to confirm whether the patient has heart disease. Initially, the hospital shares the training dataset and model parameters with the CSP in an encrypted form. CSP uses both the training dataset and model parameters to train the model. A gradient-based logistic regression technique is used in the proposed method. While training the model, the CSP updates the weight vector accordingly to reduce loss or achieve more accuracy. The CSP shares the updated weight with the hospital. Subsequently, the hospital shares the test dataset with the CSP. The CSP uses the existing trained model against the test dataset and then returns the target value to the hospital in cipher form. Later, the hospital decrypts the received prediction results using their private keys. Subsequently, the hospital communicates it to the patient in cipher form by encrypting the target value using the patient's public key. Finally, the patient uses their private key to decrypt and knows about his health status of does he has a heart problem or not. Figure 2 shows the detailed architecture of the proposed technique, which consists mainly of four algorithms: key generation and CKKS-FHE. Encryption, CKKS-FHE Decryption, and heart disease prediction using logistic regression of Encrypted Data.

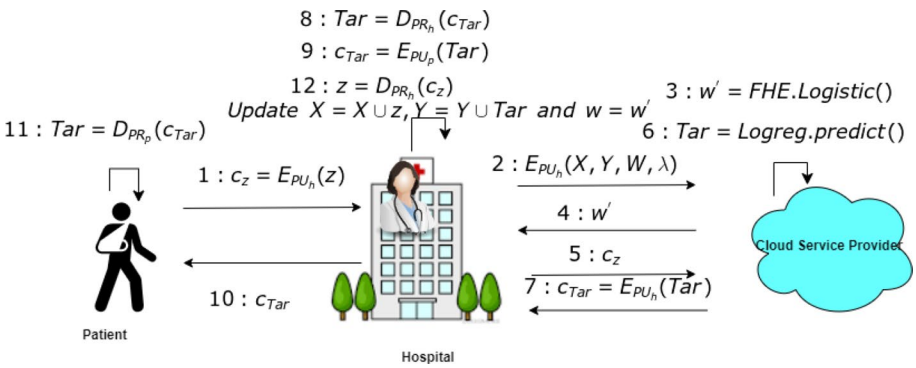


Fig. 2 Block Diagram of the proposed model

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Key concepts and polynomials in CKKS

i. Polynomial Ring *R* and Cyclotomic Polynomial:
- Let N be the degree of the polynomial, chosen as a power of 2.
- Define $M = 2N$, and the cyclotomic polynomial $\varphi_M(X) = X^N+1$.
- The polynomial ring R is defined as $Z[X]/\phi_M(X)$, which means that polynomials in this ring are considered modulo $X^N + 1$.
- The quotient ring R_q is $Z_q[X]/(X^N + 1)$, where operations are performed modulo both q (a large prime) and $X^N + 1$.
ii. Canonical Embedding Function γ and Mapping π:
- The canonical embedding function γ maps elements from R to a subset of a higher-dimensional space H.
- The mapping $\pi : H \rightarrow C^{N/2}$ transforms elements from the higher-dimensional space H to the complex vector space $C^{N/2}$.
iii. Message Polynomial and Encoding.
- A message z in CKKS is initially represented as a vector in $C^{N/2}$.
- To encode this message into a polynomial format suitable for encryption, the following steps are performed:
- The message vector $z \in C^{N/2}$ is first expanded to H by applying $\pi^{-1}(z)$.—It is then multiplied by a scaling factor Δ and rounded to the nearest integer to form r.
- The polynomial m is obtained by applying $\gamma^{-1}(r)$, resulting in the message in polynomial format.

Let *N* be the degree of the polynomial, a random number chosen as the power of *2*, then $M=2N$ and $\varphi_M(X), = X^N+1$.

Let $R = Z[X]/\varphi_M(X)$, R_q=R/R_q=$Z_q[X]/(X^N + 1)$.

Let γ be a canonical embedding function that maps R to $\gamma(R) \subseteq H$, and.

$\pi : H \rightarrow C^{\frac{N}{2}}$. A detailed description of all four algorithms is provided below.

Let N be the degree of the polynomial, a random number chosen as the power of 2, then $M=2N$ and $\varphi(M) = X^N+1$. Let $\mathfrak{R} = \mathbb{Z}[X]/\varphi_M(X)$, $\mathfrak{R}_q$=$\mathfrak{R}/\mathfrak{R}_q$=$\mathbb{Z}_q[X]/(X^N + 1)$. Let γ be a canonical embedding function that maps $\mathfrak{R}$ to $\gamma(\mathfrak{R}) \subseteq \mathcal{H}$, and $\pi : \mathcal{H} \rightarrow \mathbb{C}^{\frac{N}{2}}$. A detailed description of all four algorithms is provided below.

a. *Key generation* The patient/hospital generates public and private keys during this phase. Any patient who wants to send their medical record to a hospital located remotely securely uses this algorithm to define their key pairs, which encrypts and decrypts the messages to achieve data confidentiality. This function takes polynomial degree (N) as the argument and returns public key (PU_k) and private key (PR_k) pairs.

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Algorithm 1 Keygen ()

Input: N Output: PR_k and PU_k
Choose $PR_k \epsilon \Re_q$, a secret polynomial. Choose a uniform random polynomial a$\epsilon\Re_q$ Choose a small noisy polynomial e$\epsilon\Re_q$ compute PR_k = (-a* PR_k+e) return (PR_k, PU_k)

b. Encryption In this phase, the patient encrypts his medical record to securely send sensitive health information to a cloud server. Initially, the user decides on medical records (treated as a test dataset) and encrypts using his public key to get an encrypted dataset. The detailed algorithm for CKKS-FHE encryption is as follows. It takes medical record (‡) and PU_k as input and returns (c_1, c_2) as ciphertext. The complete encryption process comprises the encoding and encryption processes. In the encoding process of a medical record, $‡\epsilon\mathbb{C}^{N/2}$ is converted into a message polynomial; initially, $‡\epsilon\mathbb{C}^{N/2}$ is expanded to $\mathcal{H}$ by applying $\pi^{-1}(‡)$, then multiplied by the scaling factor Δ and later rounded to the nearest integer. Afterwards, it calculates γ^{-1} message in polynomial format (m). Then m is added to PU_k to get the ciphertext (c_1, c_2). Similarly, doctors/researchers use this algorithm to securely transfer the model parameters using the patient's public key.

Algorithm 2 CKKS-FHE.Encrypt()

Input: $z \in \mathbb{C}^{N/2}$,PU_k Output: ciphertext $(c_1, c_2) \in R_{q_L}{}^2$
r← $\lfloor \Delta.\pi^{-1}(z) \rfloor$ m ← $\gamma^{-1}(r)$ (c_1, c_2) ← PU_k +(m, 0)=(m-a.PRk+e,a) c_1 ← m-a.PR_k+e c_2 ←a Return (c_1, c_2)

c. Logistic regression over encrypted data The cloud server initiates this phase immediately after receiving the encrypted patient medical records. After receiving the test data, we used the existing trained model. It formulates a model that can perform both forwarded encrypted and back-propagated data to update weights, and thus train the encrypted logistic regression model on encrypted data. The detailed algorithm is as follows: We have chosen a binary cross entropy loss function with regularization of the weight vector adjusted as $\mathcal{W}_j \leftarrow \mathcal{W}_j$-$\alpha \frac{1}{P} \sum_{i=1}^{P} (\sigma(\mathcal{F}_i) - \mathcal{Y}_i)\mathcal{F}_{ij}$+0.059 $\mathcal{W}_j$.

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Algorithm 3 Logistic_Encrypted_Data()

Input: $\mathcal{F} \in \mathbb{R}^{PXD}, \mathcal{Y} \in \mathbb{R}^{PX1}$
Output: $\mathcal{W} \in \mathbb{R}^{D}$

Initialize weight vector: $\mathcal{W} \leftarrow$Enc(0)
For *iter* 0 to T, do
For i in 0, P do
temp← FHE.innerprod($\mathcal{F}_i, \mathcal{W}$)
$U_i \leftarrow FHE.Evalpoly(\sigma, \mathcal{F}_i)$
$U_i \leftarrow FHE.sub(\mathcal{F}_i, \mathcal{Y}_i)$
End for
For j in 0, D do
pp← $\frac{1}{P}\sum_{i=1}^{P} FHE.prod(U_i, \mathcal{F}_{ij})$
$g_j \leftarrow$ FHE.add (pp,0.059$\mathcal{W}_j$)
$\mathcal{W}_j \leftarrow FHE.sub(\mathcal{W}_j, FHE.prod(\alpha, g_j))$
End for
End for

Algorithm 4 FHE.add()

Input: polynomials $c_1, c_2 \in \mathfrak{R}_q^{\ 2}$
Output:c_{add}

c_{add}=$(c_1 + c_2)$ mod q_L

The routines used for FHE.add, FHE.sub, and FHE.prod are as follows.

Algorithm 5 FHE.sub()

Input: polynomials $c_1, c_2 \in \mathfrak{R}_q^{\ 2}$
Output:c_{sub}

c_{sub}=$(c_1 - c_2)$ mod q_L

Algorithm 6 FHE.prod()

Input: polynomials $c_1, c_2 \in \mathfrak{R}_q^{\ 2}$
Output:c_{mul}

for $c_1 \leftarrow (b_1, a_1), c_2 \leftarrow (b_2, a_2)$
Compute $d_0 \leftarrow b_1 b_2 \bmod q_L$, $d_1 \leftarrow (a_1 b_2 + a_2 b_1) \bmod q_L$ and $d_2 \leftarrow a_1 a_2 \bmod q_L$
$c_{mul} \leftarrow (d_0 d_1) + (P^{-1} d_2 EV_k) \bmod q_L$

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d.d. *Decryption* After receiving the predicted value from a cloud server in cipher form, a patient uses this algorithm to extract the patient's health target value, confirming whether he/she has a heart problem. The decryption process primarily comprises of two operations: decryption and decoding. Overall description of the CKKS-FHE. The decryption algorithm is described as follows: This algorithm takes ciphertext and his private key as the input and returns the predicted health status (‡). After reception of (c_1, c_2) patient multiplies c_2 with his private key and then, added to c_1, gets m $\approx c_1 + c_2.PR_k = m + e$. Subsequently, canonical embedding on m was applied to obtain s and divided by the scaling factor. Finally, to obtain the message vector ‡, he projects the s vector using π.

Algorithm 7 CKKS-FHE.Decryption()

Input: PR_k, $(c_1, c_2) \in R_{q_L}^2$ Output: $z \in \mathbb{C}^{N/2}$
m$\leftarrow c_1 + c_2.PR_k$ $s \leftarrow \Delta^{-1}(\gamma(m))$ $z \leftarrow \pi(s)$ Return (z)

Security analysis

This section introduces mathematical frameworks for various privacy-related attacks on logistic regression and suggests homomorphic encryption (HE)-based defenses against these threats. We begin by describing the logistic regression model and examining potential privacy risks with a focus on healthcare data, as shown in Figure 3.

Logistic regression can be described as follows: In a binary classification task, the goal is to predict a label y for a given input vector x. The model's parameters determine the likelihood of the target being either 0 or 1, which is expressed as

$$p(tar = 0/1|z, w) = \sigma(w^T z) \tag{6}$$

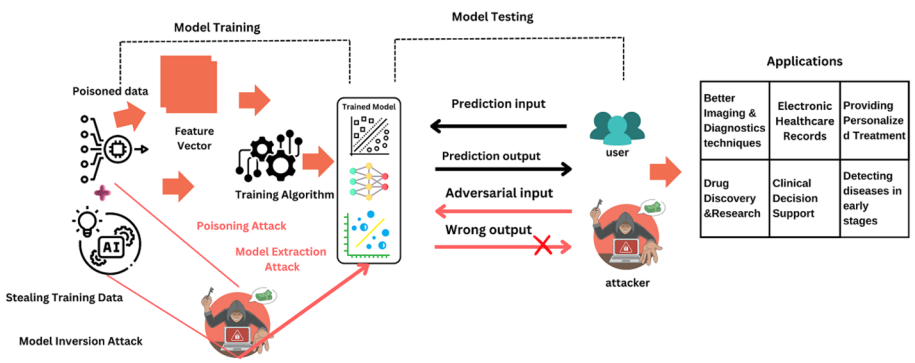


Fig. 3 Attacks on Privacy Preserving Logistic Regression

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where $\sigma(z) = \frac{1}{1+expexp(-z)}$ is the sigmoid function, and w is the weight vector learned during training.

Mathematical model for poisoning attacks on logistic regression

Consider z as an $n \times m$ matrix that represents a training dataset containing n samples, each with m m features. Let tar be a vector of n-dimensions that denotes the binary classifications for these samples, where each element is either 0 or 1.

Assume an attacker has access to the training dataset $\{(z_1, tar_1), (z_2, tar_2), \ldots, (z_n, tar_n)\}$ employed to train the logistic regression algorithm. The objective of the adversary is to alter a portion of the training dataset to maximize the loss function of the algorithm, which is defined as

$$\mathrm{L(w)} = -\sum_{\mathrm{i}} \mathrm{tar_i log}(\sigma(\mathrm{w^T z_i})) + (1 - \mathrm{tar_i})\mathrm{log}(1 - \sigma(\mathrm{w^T z_i})) \tag{7}$$

An attacker can introduce malicious examples $\{z\prime_m, tar'_m\}$ into the training dataset or perturb existing examples $\{z_i, tar_i\}$ to create new examples $\{z\prime_i, tar'_i\}$. The altered samples are created by introducing a minor modification Δz_i to the original input z_i, such that:

$$z'_i = z_i + \Delta z_i \tag{8}$$

The objective of the adversary is to discover a group of modified examples that will lead the model to incorrectly classify specific instances during the evaluation. This challenge is expressed as an optimization task.

$$Maximize L(w) \tag{9}$$

$$subject to tar'_i = tar_i \, for \, a \, subset \, of \, the \, examples$$

where $tar_{i\prime}$ is the label assigned to the perturbed example $z\prime_i$.

HE-Based Solutions to Defend against Poisoning Attacks.

For logistic regression, the weight vector w is determined by the model through the minimization of the logistic loss function:

$$\mathrm{L(w)} = -\frac{1}{\mathrm{n}}\sum_{\mathrm{i}}(\mathrm{tar_i log(p_i)} + (1 - \mathrm{tar_i})\mathrm{log}(1 - \mathrm{p_i})) \tag{10}$$

where $p_i = \frac{1}{1+exp(-z_i^T w)}$ is the predicted probability for the i th data point.

When training data and labels are encrypted using a fully homomorphic encryption (FHE) system, calculations can be performed on the encrypted information without decrypting it.

For model training, the gradient of the logistic loss function in w.r.t w is calculated as:

$$\mathrm{Grad_w L(w)} = \frac{1}{\mathrm{n}}\sum_{\mathrm{i}}((\mathrm{p_i} - \mathrm{tar_i})\mathrm{z_i}) \tag{11}$$

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A fully homomorphic encryption (FHE) system that supports mathematical operations such as addition and multiplication enable the calculation of this gradient on encrypted information.

1. 1. Encrypt the training data matrix z and label vector tar to obtain $Enc(z)$ and $Enc(tar)$.
2. Initialize the weight vector w randomly and encrypt it to get $Enc(w)$.
3. For each training iteration:—Compute the predicted probability vector P using encrypted weights and data:

$$Enc(P) = \frac{1}{1 + exp(-Enc(z^T) \times Enc(w))} \tag{12}$$

- Compute the encrypted gradient vector

$$Enc(Grad_w L(w)) = \frac{1}{n} \sum ((Enc(P) - Enc(tar)) \times Enc(z)) \tag{13}$$

- Decrypt the gradient vector to obtain the actual gradient.
- Update the weight vector w using the decrypted gradient.
- Encrypt the updated weight vector.

After the training is finished, the encrypted weight vector $Enc(w)$ can be decrypted to obtain the trained logistic regression model. By using encrypted data throughout all calculations, including the weights, any unauthorized changes to either the training data or the weights are prevented without access to the decryption key. This method effectively protects against poisoning attacks.

Mathematical model for evasion attacks on logistic regression

During an evasion attack, malicious actors manipulate input data to deceive a model's predictions while maintaining an appearance of legitimacy. This scenario assumes the attacker can access both the trained model and the test dataset$\{(z'_{test1}, tar'_{test1}), \ldots, (z\prime_{testm}, tar'_{testm})\}$.. The primary objective of an attacker is to modify the input vectors within the test dataset to cause the model to make incorrect predictions.

The attacker modifies an input vector z'_i by adding a perturbation Δz_i:

$$z\prime_i = z_i + \Delta z_i$$

The perturbation Δz_i is chosen to minimize the distance between z_i and $z\prime_i$ while ensuring the model misclassifies the input. This is formulated as:

$$minimize||\Delta z_i||subjecttof(z\prime_i) \neq tar_i \tag{14}$$

where $f(z\prime_i)$ is the predicted label for $z\prime_i$. The objective is to minimize perturbation while ensuring misclassification.

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He-Based solution to defend against evasion attacks

It is essential to demonstrate that performing homomorphic operations on encrypted data does not compromise the accuracy of the logistic regression model. The decrypted output should remain precise even if an adversary tampers with encrypted input data.

Suppose an attacker modifies the encrypted input vector $Enc(z)$ by adding a perturbation $Enc(p)$:

$$Enc(z_{new}) = Enc(z) + Enc(p) \tag{15}$$

The attacker aims to change the decrypted output tar_{new} by carefully choosing $Enc(p)$. Assume the perturbation vector is bounded by $||Enc(p)|| \leq \varepsilon$, where ε is small. In evasion attacks, this assumption restricts the alterations that an attacker can implement.

The original logistic regression output tar is:

$$tar = \sigma(w^T z + b) \tag{16}$$

The modified output tar_{new} is:

$$tar_{new} = \sigma(w^T (z + p) + b) \tag{17}$$

We extend the equation by applying the dot product's linear property and leveraging the characteristics of the sigmoid function tar_{new} as:

$$tar_{new} = \sigma(w^T z + b) + w^T p \times \sigma\prime(w^T z + b) \times (1 - \sigma(w^T z + b)) \tag{18}$$

Bounding the absolute difference between tar and tar_{new}:

$$|tar_{new} - tar| = |w^T p \times \sigma\prime(w^T z + b) \times (1 - \sigma(w^T z + b))| \tag{19}$$

Given the bounds of the sigmoid function between 0 and 1:

$$|tar_{new} - tar| \leq ||w|| \times ||Enc(p)|| \times ||\sigma\prime|_{\infty} \tag{20}$$

With $||Enc(p)|| \leq \varepsilon$:

$$|tar_{new} - tar| \leq ||w|| \times \varepsilon \times ||\sigma\prime|_{\infty} \tag{21}$$

This demonstrates that the absolute variance between the decrypted outputs tar and tar_{new} is limited by a small quantity determined by the norm of the weight vector and maximum absolute derivative value of the sigmoid function. By carefully choosing these parameters and ensuring that the perturbation vector remains sufficiently small, we can maintain the accuracy of the logistic regression model, even when faced with evasion attacks.

Adversarial examples and model vulnerability

Creating adversarial examples that can deceive machine learning models is achievable. Scientists have developed numerous techniques to generate these subtle

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alterations, which are typically unnoticeable to the human eye but can cause significant errors in model predictions. The complexity of producing effective adversarial examples is affected by multiple variables.

1. *Model Complexity* Consider a machine learning model M. Sophisticated models featuring intricate structures or high levels of nonlinearity may exhibit a greater resistance to simple disturbances. However, these models are still susceptible to sophisticated attacks, particularly those that utilize higher-order derivatives of the input-related loss function.
2. *Model Type* Different models, such as neural networks (N) or decision trees (T), exhibit varying levels of susceptibility to adversarial attacks. For instance, neural networks are often more vulnerable to gradient-based attacks, such as the Fast Gradient Sign Method (FGSM) given by

$$x' = x + \varepsilon \cdot sign(\nabla_x J(\theta, x, y)), \tag{22}$$

   where x is the original input, $J(\theta, x, y)$ is the loss function, θ are the model parameters, and y is the true label.
3. *Input Data* The vulnerability also depends on the nature of the input data $x \in R^d$. For certain datasets, small perturbations can significantly alter the decision boundary, rendering the data more susceptible to adversarial attacks.
4. *Attack Knowledge* The effectiveness of an adversarial attack often depends on the attacker's knowledge of the model, denoted as $K(M)$. For white-box attacks, in which the attacker has full knowledge of the model architecture and parameters, generating effective adversarial examples is generally easier.

Mathematical model for membership inference attacks on logistic regression In a membership inference attack, an attacker aims to determine whether a specific individual was part of the training dataset $D = \{(z_i, y_i)\}_{i=1}^n$ used to train a logistic regression model.

Let M_{LR} denote the trained logistic regression model, which outputs the probability $p(y = 1|z; w) = \sigma(w^T z)$, where $\sigma(z) = \frac{1}{1+e^{-z}}$ is the sigmoid function, and w is the weight vector.

Assume an attacker has access to the model and a set of queries $Q = \{z_{1\prime}, z_{2\prime}, \ldots, z_{m\prime}\}$, where each $z_{i\prime}$ represents an individual's information. The attacker's goal is to determine whether a specific data point $z_{i\prime}$ was used in training.

To defend against such attacks using homomorphic encryption (HE), the training data D and the weight vector w can be encrypted using Fully Homomorphic Encryption (FHE). The encrypted gradient descent update rule is given by:

$$Enc(w') = Enc(w) - \alpha \cdot Enc(Z^T) \cdot (Enc(y) - Enc(y')), \tag{23}$$

where α is the learning rate, Z is the input matrix, and y and $y\prime$ are the true and predicted labels, respectively.

After a fixed number of iterations, we obtain the final encrypted weight vector $Enc(w^*)$, which is used to predict output values for new encrypted test data. The prediction for a new input z_{test} is:

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$$Enc(\widehat{y}) = \sigma(Enc(z_{test}^T) \cdot Enc(w^*)). \tag{24}$$

Decryption of the predicted output is performed as:

$$\widehat{y} = Dec(Enc(\widehat{y}))$$

This ensures that an attacker cannot infer membership because all computations are performed on the encrypted data. However, FHE can be computationally expensive, which makes it essential to balance privacy and performance.

Mathematical model for model inversion attacks on logistic regression In a model inversion attack, the attacker attempts to reconstruct the training data D from the logistic regression model M_{LR}.

Consider a binary classification problem where the model is trained on a dataset $D = \{(z_i, y_i)\}_{i=1}^n$. Given access to the model and its predictions $\{z_{1\prime}, z_{2\prime}, \ldots, z_{m\prime}\}$, the attacker computes the partial derivative of the model's output with respect to each input feature:

$$\frac{\partial p(y = 1|z, w)}{\partial z_j} = \sigma(w^T z) \cdot (1 - \sigma(w^T z)) \cdot w_j \tag{25}$$

The attacker can then solve the following optimization problem to reconstruct the original input z_i:

$$\min_{z_{i\prime}} \| z_i - z_{i\prime} \|_2 \, subjectto \, p(y = 1|z_i, w) = p(y = 1|z_{i\prime}, w) \tag{26}$$

To defend against this, homomorphic encryption can be used to encrypt the weight vector w, so that:

$$Enc(p) = \sigma(Enc(z^T) \cdot Enc(w)),$$

and noise can be added to the predicted probability:

$$Enc(p_{noisy}) = Enc(p) + Enc(\eta),$$

where η is a random noise term. This makes it difficult for the attacker to accurately reconstruct the input data.

Mathematical model for model extraction attacks on logistic regression In a model extraction attack, the attacker aims to replicate a trained logistic regression model M_{LR} without access to the original training data D.

Assume the model is trained on $D = \{(z_i, y_i)\}_{i=1}^n$, and the attacker has access to the model and a set of input–output pairs $\{(z_{1\prime}, y_{1\prime}), (z_{2\prime}, y_{2\prime}), \ldots, (z_{m\prime}, y_{m\prime})\}$ generated by querying the model. The attacker attempts to solve the following optimization problem to extract the weight vector w:

$$\min_{w\prime} \| w - w\prime \|_2 \, subjectto \, p(y = 1|z', w') = y', \tag{27}$$

where w' is the extracted weight vector.

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Table 2 CKKS Homomorphic parameters

poly_modulus_degree	4096	8192
coeff_mod_bit sizes	[40, 20, 40]	[40, 21, 21, 21, 21, 21, 21, 40]
ctx_training.global_scale	2^{20}	2^{21}

Table 3 User Computation time

	Function	Mathematical Operations
Patient	KG, ETDS, DCR	KG: 1Padd,1Psub,1Pmul ETDS: D*(2Inv,2Pmul,1Psub,1Padd) DCE: 4Pmul,1Inv,1Map,1Padd
Hospital	KG, ETRDS, DCR, ECR	KG: 1Padd,1Psub,1PMul ETRDS: P*D*(2Inv,2Pmul,1Psub,1Padd) DCR:4Pmul,1Inv,1Map,1Padd ECR:2Inv,2PMul,1Psub,1Padd
Server	HELR, ECR	HELR: ECR: 2Inv,2PMul,1Psub,1Padd

* *KG* Key pair Generation, *ETDS* Encrypt Test DataSet, *DCR* Decryption of Classification Result, *ETDS* Encrypt Training DataSet, *ECR* Encryption of Classification Result, *HELR* Homomorphic Logistic Regression

To defend against this, homomorphic encryption can be applied to encrypt the model parameters.

$$Enc(y) = \sigma(Enc(w^T) \cdot Enc(z)). \tag{28}$$

Because the attacker cannot decrypt the model parameters or encrypted output predictions without the decryption key, the optimization problem becomes infeasible.

$$\min_{w'} \| Enc(y) - Enc(\sigma(w'^T z)) \|_2, \tag{29}$$

In conclusion, homomorphic encryption offers strong protection against various attacks on logistic regression models, including membership inference, model inversion, and model extraction. However, this approach results in a substantial increase in computational costs. When implementing these methods, it is crucial to strike a balance among privacy protection, security measures, and practical computational requirements.

Experimentation and results

To develop the CKKS encryption and decryption scheme, the proposed technique was implemented using the PyTorch and TenSEAL libraries on Google Colab. Table 2 lists the parameters used in the CKKS. The key pair required for dot product operations was generated using the ctx_eval.generate_galois_keys() method. A logistic regression model was built using Gradient Descent with a learning rate of 1 and a binary cross-entropy loss function. The number of epochs varied between 5, 10, 20, and 50, with updates to the weight vector accordingly.

In this study, we analyzed the performance of two algorithms: logistic regression without data encryption and a logistic regression (LR) model on encrypted data using homomorphic encryption with a polynomial modulus of degrees 4096 and 8192 for heart disease prediction. The Heart dataset (https://www.kaggle.com/datasets/johnsmith88/

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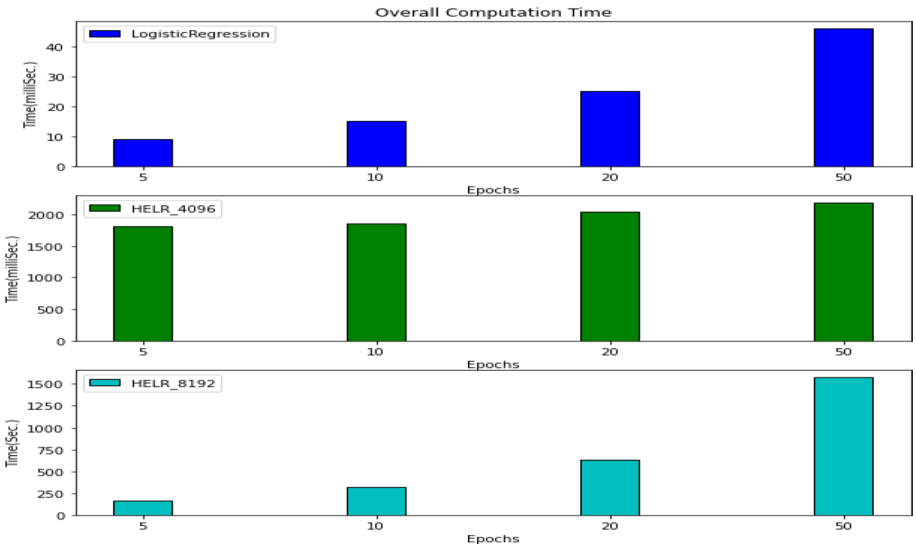


Fig. 4 Overall computation Time

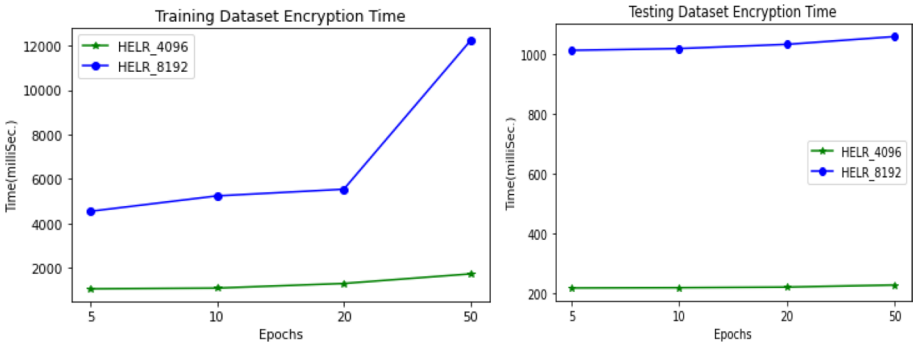


Fig. 5 Training and Testing Dataset Encryption Time

heart-disease-dataset) [27] was used for the analysis. The parameters used in our proposed method are presented in Table 2, and their performances were compared in terms of the overall computation time. Table 3 shows the total number of patients, hospitals, and CSP operations performed. Only the computation time for the different operations, excluding the communication cost, was considered. The proposed method requires the patient to perform key generation (KG), medical record encryption (ETDS), and decryption of classification results (DCE). The user assumes three polynomials: one polynomial addition (Padd), one polynomial subtraction (Psub), and one polynomial multiplication (PMul) to generate the key pair. The detailed mathematical operations for encryption and decryption are shown in the same table, where the inverse and mapping operations are represented as Inv and map, respectively. Figure 4 illustrates the overall computation time, indicating that logistic regression on plain data requires milliseconds to complete the prediction, whereas logistic regression on encrypted data takes longer. Furthermore, the computation time increases with the degree.

Figure 5 shows a comparative analysis of the computation time to perform the training dataset and testing dataset encryption time using modulus polynomials 4096 and

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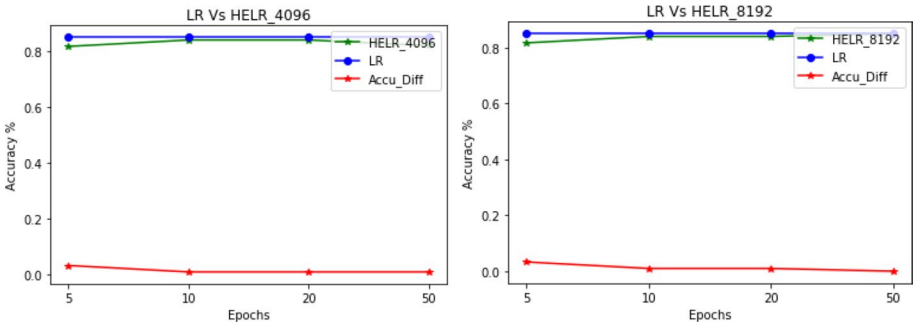


Fig. 6 LR Vs HELR Accuracy

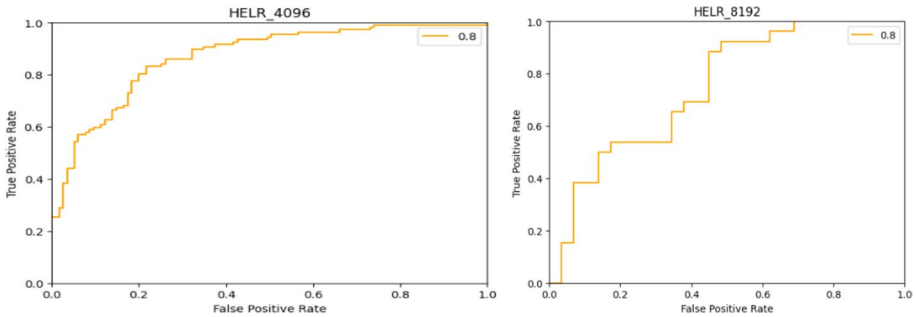


Fig. 7 HELR ROC Curve

8192. It was observed that 8192 required more computation time than 4096. The performances of both techniques were compared by changing the number of epochs to 5, 10, 20, and 50. HELR_4096 has taken 1061, 1093, 1301, and 1731 ms to encrypt the training dataset, whereas HELR_8192 has taken 4550,5242, 5542, and 12225 ms. In addition, we can observe that with an increase in epoch values, the computation time also increases. Similarly, HELR_4096 took 219, 220, 222, and 229 ms to encrypt the test dataset, and HELR_8192 took 1013,1019, 1033, and 1059 ms, respectively.

Figure 6 shows the accuracy analysis of Logistic Regression (LR) and Logistic Regression using Homomorphic Encryption (LR_HE) using modulus polynomials 4096 and 8192. The accuracy was analyzed for 5, 10, 20, and 50 epochs. It was observed that accuracies of 85%, 84%, and 84% were achieved in LR, HELR_4096, and HELR_8192, respectively. From the accuracy graphs, the accuracies of LR_HELR_4096 and HELR_8192 are approximately equal to that of LR, with a deviation of approximately 1%. In addition, we observed that the performance remained the same irrespective of the change in epoch count for all three techniques.

The area under the ROC curve (receiver operating characteristic curve) indicates the performance of the classifier. The ROC curve was constructed by plotting the (FPR and TPR) pairs for each possible threshold value. The values of TPR and FPR were computed by varying the threshold values of 0.2, 0.4, 0.6, and 0.8. The ROC was constructed by plotting (FPR, TPR) for all threshold values, and observed that at 0.8 thresholds, classifiers performance was better. Figure 7 shows the ROC curves for HELR_4096 and HELR_8192 at a threshold of 0.8, with an optimal AUC. From the figure, it can be

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Table 4 Comparison of the proposed method with existing framework

Author	Objective	Dataset	Model	Advantages	Limitations
X. Zhao et al., 2022	secure multi-party computation protocol	COVID-19 negative X-ray image dataset samples	BFV homomorphic encryption. Deep Learning	Efficient for Integer Arithmetic operations. Good performance for operations involving integers	Less Flexibility for Real-World Applications. Less Efficient for Approximate Arithmetic
[24]	vertically partitioning dataset by asynchronous gradient sharing	MNIST Dataset	CKKS algorithm. Logistic regression	Efficient for Small Datasets Strong security guarantees against various cryptographic attacks	Struggle to handle the high dimensionality of image data efficiently. High computational overhead
proposed	Patient centric data privacy through CKKS technique	Heart Disease Dataset	CKKS algorithm. Logistic regression	Strong security guarantees against various cryptographic attacks Efficient for Small Datasets. Compatibility with Real-World Applications	Scalability of logistic regression with HE for large healthcare datasets

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observed that with a higher AUC value, the performance of HELR_4096 is better than that of HELR_8192. Comparisons of the proposed method with existing frameworks are presented in Table 4.

The Fig. 8 compares the performance of two methods, HELR and HESVM, across three datasets (heart, statlog (https://www.kaggle.com/datasets/shubamsumbria/statlog-heart-data-set) [36] and (https://www.kaggle.com/datasets/shubamsumbria/statlog-heart-data-set),and Framingham(https://www.kaggle.com/datasets/aasheesh200/framingham-heart-study-dataset)) [35] under two parameter configurations (4098 and 8192). For the heart dataset, HELR consistently outperforms HESVM, achieving a score of 81.7 with the 4098 configuration and improving to 85 with the 8192 configuration, compared to HESVM's scores of 67 and 79, respectively. Similarly, for the statlog dataset, HELR achieves perfect scores of 100 under both configurations, while HESVM shows improvement from 69 with the 4098 configuration to 88 with the 8192 configuration but still falls short of HELR's performance. In contrast, the results for the framingham dataset are mixed. With the 4098 configuration, HESVM slightly outperforms HELR (scores of 63 vs. 62), but HELR takes the lead with the 8192 configuration, achieving 65 compared to HESVM's 61.

Overall, HELR demonstrates superior performance across most datasets and configurations, particularly excelling in the statlog dataset where it achieves perfect scores. Both methods generally show improved performance with the 8192 configurations, indicating the potential benefit of higher parameter settings. However, HESVM's performance remains consistently below HELR for the heart and statlog datasets, while the results for framingham suggest a more competitive but still slightly weaker performance by HESVM.

Comparative analysis

The proposed scheme was evaluated against two existing schemes that utilize Homomorphic Encryption (HE) with Logistic Regression (LR), despite their different intended purposes and methodological variations. Table 4 presents a comparison. In 2022, X. Zhao et al. introduced a privacy-preserving federated learning system for medical data using homomorphic encryption. Their approach uses a secure multiparty computation protocol to protect the deep learning model from potential threats. The BFV

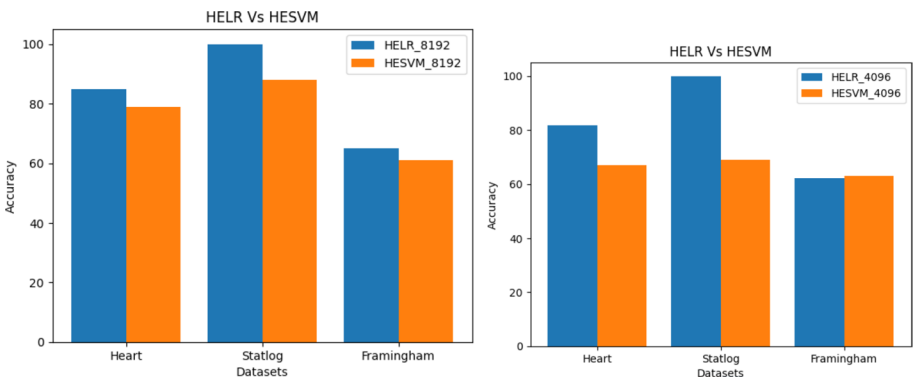


Fig. 8 Performance analysis of Logistic Regression vs SVM on three datasets (heart, statlog, and Framingham)

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homomorphic encryption technique used in this method is primarily suited to integer operations, which is a limitation.

[24] addressed the pressing need for efficient and secure multi-party training using vertically partitioned datasets. They focused on developing efficient training methods for logistic regression models. Their protocol employed logistic regression modelling of vertically partitioned data through asynchronous gradient sharing. The authors emphasized the vertical partitioning of the training data, encrypting it using the CKKS algorithm, and applying logistic regression for model training. However, their study utilized an image dataset to predict COVID-19 negative X-ray image dataset samples.

Our study focused on a heart disease dataset by employing the CKKS technique to enhance privacy protection. CKKS, an encryption method primarily utilized in homomorphic encryption, enables computations of encrypted information without prior decryption. This scheme offers numerous benefits, including versatility with respect to data types, support for arbitrary precision, and interoperability. As a result, CKKS is well suited for a diverse datasets and range of practical applications that require secure computations on encrypted data.

Results and interpretation

1. *Computation Time* Encryption time increased with higher polynomial sizes and epoch counts. The HELR model using 4096 was computationally more efficient than 8192, making it suitable for resource-limited environments. However, 8192 offers stronger security, which may justify its higher overhead in critical applications.
2. *Accuracy* HELR models achieved an accuracy of 84%, closely matching the baseline Logistic Regression (LR) model's 85%. This demonstrates that the homomorphic encryption process does not significantly compromise predictive accuracy, maintaining its utility in privacy-preserving scenarios.
3. *Security* ROC analysis revealed that HELR_4096 had a higher AUC than HELR_8192, indicating better performance under certain configurations. This suggests that smaller polynomials may balance security and efficiency effectively in specific contexts.
4. *Comparative Performance* Across datasets, HELR consistently outperformed HESVM, particularly in the statlog dataset, where it achieved perfect scores. The results demonstrate HELR's robustness and suitability for diverse data types and parameter configurations.

Interpretation

The findings confirm that the HELR framework offers a practical balance between privacy, computational efficiency, and predictive accuracy. While higher polynomial sizes enhance security, the trade-off in computational time should guide configuration choices based on application needs. HELR's superior performance across datasets underscores its potential for secure and effective healthcare predictions.

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The results confirm the HELR model's effectiveness in ensuring privacy while maintaining high predictive performance.

Conclusion and future work

In this study, we proposed an HE-aware logistic regression model built on encrypted data using CKKS. The proposed method can be used to outsource the training phase of logistic regression to a cloud service in a privacy-preserving manner. We conducted experiments using the Tenseal package on real datasets from Kaggle to evaluate the performance of both the logistic regression (over unencrypted data) and the proposed system. We conducted a security analysis to show that the proposed system can defend against poison, evasion, member inference, model inversion, and model extraction in the respective stages of machine learning. The performance of the proposed system provided satisfactory results relative to the model without HE, with a minimal difference in accuracy between 0.01 and 0.03. As a future scope, one can investigate and explore extension HELR framework to multi-class classification problems and propose HE-aware multi class healthcare prediction systems.

Author contributions
First Author (V S N): Contributed to the design and conceived the original and supervised the project. Entire drafting of the paper. Second Author (SR): Contributed to the implementation of the research, and analysis of the results and to the writing of the manuscript.

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Availability of data and materials
No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to Participate
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Competing interests
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